

RESEARCH ARTICLE



Propranolol enhances the oncolytic effect of newcastle disease virus on canine mammary tumor cell by modulating the IFN-I-mediated JAK-STAT signaling pathway

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ABSTRACT

Canine mammary tumors (CMT), the most common neoplasms in intact female dogs, lack effective treatments for advanced cases. Newcastle disease virus (NDV), an oncolytic virus, kills tumor cells directly and stimulates antitumor immunity. However, NDV also activates Type I interferon (IFN-I) signaling, which restricts its replication via the JAK-STAT pathway. This study investigated whether propranolol, known to suppress antiviral responses, could enhance NDV's oncolytic effect in CMT. *In vitro*, using the CMT-U27 cell line, the combination reduced cell viability, migration, and invasion. Propranolol increased NDV replication and suppressed NDV-induced IFN-I release and JAK-STAT signaling pathway activation. In the CMT-U27 xenograft model, both NDV alone and the combination therapy prolonged the survival and suppressed tumor growth, with no significant difference between the two. However, the combination markedly enhanced intratumoral NDV replication by inhibiting IFN-I production and the JAK-STAT signalling pathway. These findings indicate that propranolol enhances NDV-mediated oncolysis in CMT by inhibiting the IFN-I/JAK-STAT pathway, increasing viral load, and represents a promising combined therapeutic strategy.

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1. Introduction

Oncolytic therapy utilising Newcastle disease virus (NDV) has demonstrated significant advances in recent years, with evidence indicating NDV-mediated antitumor effects through multiple pathways (Schneider and Chevillotte, 2014; Li et al., 2022; Schirrmacher and van Gool, 2022; Han and Wu, 2023; Yang et al. 2024). Within this regulatory network, Type I interferon (IFN-I), key mediators of the host immune system, play a central role in tumour suppression. NDV stimulates tumour cells to produce IFN-I (Schneider and Chevillotte, 2014; Han and Wu, 2023), which in turn activates both anti-inflammatory and anti-tumour responses. IFN-I binding to IFNAR1/IFNAR2 receptors triggers JAK-STAT signalling pathway activation, enhancing antiviral immunity and inhibiting NDV replication within tumours (Santos et al., 2021; Guttman-Yassky et al. 2023; Zanin et al. 2023). Thus, the dynamic balance between IFN-I-mediated antiviral responses and the oncolytic activity of NDV represents a critical regulatory axis in NDV-based tumour therapy.

Propranolol is a non-selective β -adrenergic receptor antagonist and is clinically employed in the management of hypertension, pheochromocytoma, myocardial infarction, arrhythmia, angina pectoris, and hypertrophic cardiomyopathy (Abdulrahman et al. 2017). Beyond its cardiovascular applications, propranolol has also been employed as an adjunctive treatment for various tumours (Puzderova et al., 2023; Lin et al. 2023; Wang et al. 2023; Terauchi and Fujii, 2023), including epidermoid carcinoma (Kelly and Butcher 1975), malignant insulinoma (Blum and Rusecki, 1983), lung adenocarcinoma (Schuller and Cole 1989), and breast cancer (Slotkin and Zhang, 2000). Notably, Hu et al. reported that propranolol enhanced

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the replication of oncolytic herpesvirus in tumours by inhibiting STAT3 activation, thereby dampening the host antiviral response (Hu et al., 2021). Furthermore, propranolol has been investigated in combination with the oncolytic virus T1012G as a potential therapeutic strategy for gastric and colorectal carcinoma (Hu et al., 2021, Hu et al., 2021).

The objective of this study was to evaluate whether propranolol enhances the oncolytic efficacy of NDV in CMT-U27 cells by modulating the IFN-I-mediated JAK-STAT signalling pathway and to assess the antitumor potential of this combination both *in vitro* and in the CMT xenograft mouse model.

2. Materials and methods

2.1. Cells and virus

The canine mammary tumour cell line CMT-U27 (CRL-3456) was obtained from the American Type Culture Collection (ATCC, Manassas, USA) and cultured in Roswell Park Memorial Institute (RPMI-1640) medium (L220KJ, Basal Media, Shanghai, China) supplemented with 10% foetal bovine serum (FBS; F101-01, Vazyme, Nanjing, China) and 1% Penicillin-Streptomycin-Gentamicin Solution (C0223, Beyotime, Beijing, China). Madin-Darby canine kidney (MDCK) cells and African green monkey kidney epithelial (Vero) cells were maintained in Dulbecco's Modified Eagle Medium/Nutrient Mixture F-12 (DMEM/F12; L310KJ, Basal Media, Shanghai, China) containing 10% FBS. MDCK cells were used as a non-neoplastic control, and Vero cells were employed for virus titration. All cell lines were incubated at 37 °C in a humidified incubator with 5% CO₂. Cell morphology and growth characteristics were routinely monitored.

NDV (LaSota strain) was propagated in 10-day-old specific-pathogen-free (SPF) chicken embryos. At 72-hour post infection (h.p.i), the allantoic fluid was collected and centrifuged at 3,000 g for 20 min at 4 °C. The supernatant was filtered through a 0.22 µm membrane filter, and stored at -80 °C for further analysis. Virus titre were obtained as the median tissue culture infectious dose per mL (TCID₅₀ mL⁻¹) using the Reed and Muench method. For titration, Vero cells were seeded at 3,000 cells per well in 96-well plates. Once the cells reached approximately 70–80% confluency, they were exposed to varying concentrations of the virus. After a 2 h incubation period, 100 µL of medium supplemented with 2% FBS was added to each well. Cytopathic and morphologic effects were evaluated daily for three days (Al-Ziaydi et al. 2020).

2.2. Cell viability assay

CMT-U27 cells were seeded into a 96-well plate at a density of 2,500–3,000 cells per well. Propranolol (HY-B0573, MedChemExpress, New Jersey, USA) was diluted in RPMI-1640 medium supplemented with 2% FBS to final concentrations ranging from 1 nM to 100 nM. Experimental groups included a control group and propranolol treatment group (1, 5, 10, 20, 40, 100 nM). After 48 h of treatment, Cell Counting Kit-8 (CCK-8; A311, Vazyme, Nanjing, China) reagent was added to each well. Plates were incubated at 37 °C in the dark for 30–60 min, and the optical density (OD) was measured at 450 nm using a Multiskan FC Enzyme Analyser (Thermo Fisher Scientific, Massachusetts, USA). The highest non-toxic concentration of propranolol (100 nM) was selected for use in the subsequent experiments. CMT-U27 cells were treated with 100 nM propranolol, NDV at a multiplicity of infection (MOI) of 1, or a combination of both for 48 h. Cell viability was assessed using the CCK-8 assay as described above.

2.3. Combination therapy regimen with NDV and propranolol

CMT-U27 cells were seeded into culture plates and allowed to grow until they reached 80–90% confluency. The cells were subsequently allocated into four groups: the mock-treated group (M group), the propranolol-treated group (P group), the NDV-infected group (N group), and the group subjected to combined NDV infection and propranolol treatment (NP group). The M group was maintained in RPMI-1640 medium containing 2% FBS; the P group was cultured in RPMI-1640 medium with 2% FBS and 100 nM propranolol; the N group was infected with NDV at a MOI of 1 for 2 h at 37 °C, following viral absorption, the medium was replaced with fresh RPMI-1640 containing 2% FBS; the NP

group underwent the same NDV infection protocol as the N group, after which the medium was replaced with RPMI-1640 containing 2% FBS and 100 nM propranolol. Cells were washed with Dulbecco's phosphate-buffered saline (DPBS; B210KJ, Basal Media, Shanghai, China) before and after NDV infection to remove residual virus.

2.4. Enzyme-linked immunosorbent assay (ELISA)

CMT-U27 and MDCK cells were treated with different titre of NDV, propranolol, or a combination of both. Following treatment, culture supernatants were collected and the IFN-I concentration was measured by IFN- α (YH-090615C, Yihe Biological, Shanghai, China) and IFN- β (YH-090616C, Yihe Biological, Shanghai, China) ELISA kit, according to the manufacturer's instructions.

2.5. Real-time quantitative polymerase chain reaction (RT-qPCR)

Total RNA was extracted from cells or fresh tissues using the Total RNA Extractor Trizol (R401-01, Vazyme, Nanjing, China) according to the manufacturer's instructions. HiScript III qRT SuperMix (R433, Vazyme, Nanjing, China) was used to convert the RNA to cDNA. RT-qPCR was performed on an ABI 7300 Real-Time PCR system (Thermo Fisher Scientific, Massachusetts, USA) with AceQ qPCR SYBR Green Master Mix (Q141, Vazyme, Nanjing, China). Gene expression levels were normalised to β -actin expression and expressed as fold changes in relative to the control group using the $2^{-\Delta\Delta CT}$ method. The primer design and verification are presented in the supplementary materials (Table S1-2, Figure S1-3).

2.6. Western blot analysis

Total protein was extracted from cells or tissues by lysis in radioimmunoprecipitation assay (RIPA) buffer (R0010, Solarbio, Beijing, China) supplemented with 1% phenylmethylsulfonyl fluoride (PMSF) and 2% phosphatase inhibitor (P1081, Beyotime, Shanghai, China). Protein concentrations were determined using a bicinchoninic acid (BCA) protein assay kit (ZJ101, Beyotime, Shanghai, China). Protein samples were denatured, separated by SDS-PAGE, and transferred onto 0.22 μ m polyvinylidene fluoride (PVDF) membranes (ISEQ00010, Merck Millipore, Darmstadt, Germany). Membranes were blocked with blocking buffer (AIWB-004, Affinibody LifeScience, Wuhan, China), incubated with primary antibodies overnight at 4 °C, and then incubated with horseradish peroxidase (HRP)-conjugated secondary antibodies for 2 h at room temperature. Protein bands were visualised using the enhanced chemiluminescence (ECL) system (BMU101-CN, Abbkine, Wuhan, China) and quantified with ImageJ software (v1.8.0, National Institutes of Health, Maryland, USA). The following antibodies were used: β -actin (1: 5000; AF12301, Affinibody LifeScience, Wuhan, China), JAK1 (1: 1500; T57173, Abmart, Shanghai, China), phospho-JAK1 (p-JAK1, 1: 1500; TP56310, Abmart, Shanghai, China), STAT1 (1: 1500; YP-Ab-01278, Abmart, Shanghai, China), phospho-STAT1 (p-STAT1, 1: 1500; YP-Ab-01375, UPingBio, Hangzhou, China), STAT3 (1: 1500; YP-Ab-01150, UPingBio, Hangzhou, China), PKR (1: 1500; YP-Ab-14943, UPingBio, Hangzhou, China), HRP-conjugated goat anti-rabbit IgG (H + L) secondary antibody (1: 5000; SA00001-2, ProteinTech group [PTG], Chicago, USA), HRP-conjugated goat anti-mouse IgG (H + L) secondary antibody (1: 5000; SA00001-1, PTG, Chicago, USA).

2.7. Wound-healing assay

CMT-U27 cells were plated into 12-well plates at a density of 3×10^4 cells per well and cultured until they reached 90% confluency. A linear scratch was made in the cell monolayer using a sterile 200 μ L pipette tip. Subsequently, cells were treated with NDV (MOI = 0.01), propranolol (100 nM) or a combination of both. Following treatment, cells were incubated in serum-free RPMI-1640 medium for 24 h. Cell migration was evaluated and imaged at 6 and 24 h.p.i, and the scratch area was quantified using ImageJ software.

The migration area was calculated as a percentage of the initial scratch area, using the formula: Migration area (%) = [Area (0) - Area (t)]/Area (0) $\times$ 100%.

2.8. Immunofluorescence

CMT-U27 cells were pre-seeded onto cell climbing slices (BS-18-RC, Biosharp, Hefei, China) and assigned into three groups: M group, N group and NP group. Treatments were administered as described in [Section 2.3](#). At 24 h.p.i, cells were fixed with 4% paraformaldehyde (105008, Xilong Scientific, Shantou, China), permeabilized with 0.1% Triton X-100 (ST797, Beyotime, Shanghai, China) in PBST, and blocked with 5% bovine serum albumin (BSA, Applygen Technologies Inc., P1621, Beijing, China) solution. Following blocking, cells were incubated overnight at 4 °C with the NDV-HN antibody (1:50; sc-53562, Santa Cruz, Texas, USA). Subsequently, cells were incubated at room temperature in the dark with a goat anti-mouse IgG secondary antibody (1:200; SA00001-1, PTG, Chicago, USA). Nuclei were counterstained with DAPI, and antifade mounting medium was applied. Fluorescence images were captured using a fluorescence microscope (Nikon, Tokyo, Japan) equipped with 420 nm and 590 nm excitation/emission filter sets.

2.9. Transwell assay

Matrix gel (082724 T, Mogengel Biotechnology, Xiamen, China) was diluted with cold serum-free medium and applied to the upper membranes of 24-well Transwell inserts. CMT-U27 cells (either NDV-infected or mock-infected) were resuspended in serum-free medium and seeded into the upper chambers at a density of 1×10^5 cells per well, with or without 100 nM propranolol depending on experimental groups. The lower chambers were filled with complete medium containing 10% FBS. After 7 days of incubation, the cells were fixed, stained, and visualised under an inverted microscope (MI40, Mshot, Guangzhou, China). The number of migrated cells were quantified using ImageJ software.

2.10. Oncolysis in vivo

The study was approved by the Institutional Animal Care and Use Committee of Nanjing Agricultural University (No. 20241015206). CMT-U27 cells with good growth condition were suspended with RPMI-1640 medium and mixed with matrix gel at a 1:1 ratio. The final cell concentration was adjusted to 5×10^7 mL⁻¹. Six-week-old BALB/c nude mice were subcutaneously injected with 100 µL of cell suspension into the right axilla. Tumour-bearing mice were housed under SPF conditions and monitored daily for tumour development. Within 14 days, approximately 70–80% of mice developed palpable solid tumours. Once tumours reached a volume of approximately 100 mm³ (volume = [length × width²]/2), the mice were randomly assigned into four groups ($n = 6$, per group): M group (intratumoral and intraperitoneal injection of DPBS), P group (intraperitoneal injection of propranolol at 3 mg/kg (Zheng and Du, 2024) and intratumoral injection of DPBS), N group (intratumoral injection of 100 µL NDV at 2.3×10^6 TCID₅₀ mL⁻¹ and intraperitoneal injection of 100 µL DPBS), and NP group (intratumoral injection of 100 µL NDV and intraperitoneal injection of propranolol at 3 mg/kg). Treatments were administered every other day. Body weight and tumour volume were measured everyday. Mice were euthanized via cervical dislocation if they lost more than 20% of their body weight within one week or if the tumour volume exceeded 1000 mm³. The entire experiment lasted for 15 days, after which all surviving nude mice were euthanized.

Before euthanasia, blood samples were collected via retro-orbital bleeding for ELISA analysis. A portion of each tumour was fixed in 4% formaldehyde solution for subsequent immunofluorescence analysis to assess the NDV-HN expression. The immunofluorescence procedure can be found in [Section 2.8](#). The remaining tumour tissues were stored at −80 °C for later analysis of genes and proteins related to the JAK-STAT signalling pathway.

2.11. Statistical analysis

Statistical analyses were performed with GraphPad Prism 9.0 (Dotmatics, San Diego, USA). The Shapiro-Wilk test was used to assess normality, and all data met the assumption of normal distribution. Comparisons between two groups were performed using the unpaired Student's *t*-test. Comparisons among multiple groups were analysed by one-way ANOVA followed by Tukey's post hoc test for multiple comparisons. Drug interaction between NDV and propranolol was assessed using the Bliss independence model. Survival

curves were compared using the Kaplan-Meier method with the log-rank test, and *P*-values were adjusted for multiple comparisons using the Bonferroni correction. Data are presented as mean $\pm$ SEM. All the experiments were performed in triplicate and repeated at least three times. *P*-value < 0.05 was considered a statistically significant difference. The combination index (ΔE) > 0 was considered synergistic, while $= 0$ was considered additive.

3. Results

3.1. NDV induces IFN-I release from CMT-U27 cells but not from MDCK cells

CMT-U27 cells infected with NDV exhibited time- and dose-dependent induction of IFN-I production. The IFN-I levels in the N group were significantly higher than those of the M group at 24 and 48 h.p.i. (Figure 1A, B). RT-qPCR analysis further confirmed that IFN-I gene expression was upregulated after 48 h of infection with 1 MOI NDV (Figure 1C, D). In MDCK cells infected with NDV (MOI = 1), RT-qPCR showed a significant increase in IFN-I mRNA levels at 24 h.p.i. relative to controls ($P < 0.05$), although no corresponding increase in secreted protein was detected by ELISA. By 48 h.p.i., both IFN-I mRNA and secreted protein levels in NDV-infected MDCK cells were comparable to those in controls (Figure 1E–H). These findings suggest that NDV induces IFN-I production in both CMT-U27 and MDCK cells at 24 h.p.i. However, while IFN-I levels in MDCK cells returned to baseline by 48 h.p.i., expression remained elevated in CMT-U27 cells.

3.2. NDV activates the JAK-STAT signalling pathway in CMT-U27 cells

To examine the effect of NDV on the JAK-STAT signalling pathway, CMT-U27 were infected with NDV at 1 MOI for 24 and 48 h. RT-qPCR confirmed significant increases in mRNA levels of STAT3, PKR, MX1 and OAS-1 in NDV-infected cells ($P < 0.05$) (Figure 2A, B). In MDCK cells infected with NDV, the mRNA expression of MX1 and OAS-1 was significantly reduced at 24 h.p.i. ($P < 0.05$). By 48 h.p.i., no significant differences were observed in the expression of any examined JAK-STAT pathway-related genes compared with controls (Figure 2C, D). Western blot results showed that the expression of JAK-STAT signalling pathway-related proteins including JAK1, STAT1, *p*-STAT1, STAT3 and PKR, did not change significantly at 24 h.p.i., but were significantly upregulated at 48 h.p.i. ($P < 0.05$) (Figure 2E–G). These findings suggest that activation of the JAK-STAT signalling pathway may be involved in the cellular response of CMT-U27 cells to NDV infection, potentially contributing to the observed antitumor effects.

3.3. Propranolol enhances the inhibition effect of NDV on the viability, migration and invasion of CMT-U27 cells

Treatment with 1–100 nM propranolol for 48 h did not significantly affect the viability of CMT-U27 cells ($P > 0.05$). Therefore, 100 nM, the highest non-toxic concentration, was chosen for subsequent experiments. After 48 h, cell viability in the NP group was significantly reduced compared to both the N group ($P < 0.01$) and the M group ($P < 0.001$, Figure 3A, B). The interaction between NDV and propranolol was subsequently evaluated using the Bliss independence model. The mean Bliss excess (ΔE) was 0.0019 (95% CI: -0.1102 to 0.1140). As the confidence interval included zero and the one-sample *t*-test was not statistically significant ($P > 0.05$), the interaction was consistent with Bliss additivity.

Consistently, cell migration was markedly suppressed in the NP group, where less than 30% of the wound was healed after 24 h, in contrast to almost complete closure in the M group ($P < 0.001$) and significantly lower than in the N group ($P < 0.05$). Bliss analysis yielded a ΔE value of 0.198 (95% CI: -0.149 to 0.545 , $P > 0.05$), again indicating an additive interaction between NDV and propranolol (Figure 3C, D).

A similar trend was observed in the invasion assay: both the NP and N groups showed significantly decreased invasion compared to the M group ($P < 0.001$), and the NP group demonstrated a significantly greater reduction in invasion capacity compared to the N group ($P < 0.05$). Bliss analysis yielded a ΔE value of 0.2157 (95% CI: 0.149 to 0.282 , $P < 0.001$), indicating a synergistic interaction between NDV and propranolol (Figure 3E, F). Together, these results indicate that propranolol enhances the inhibitory effects

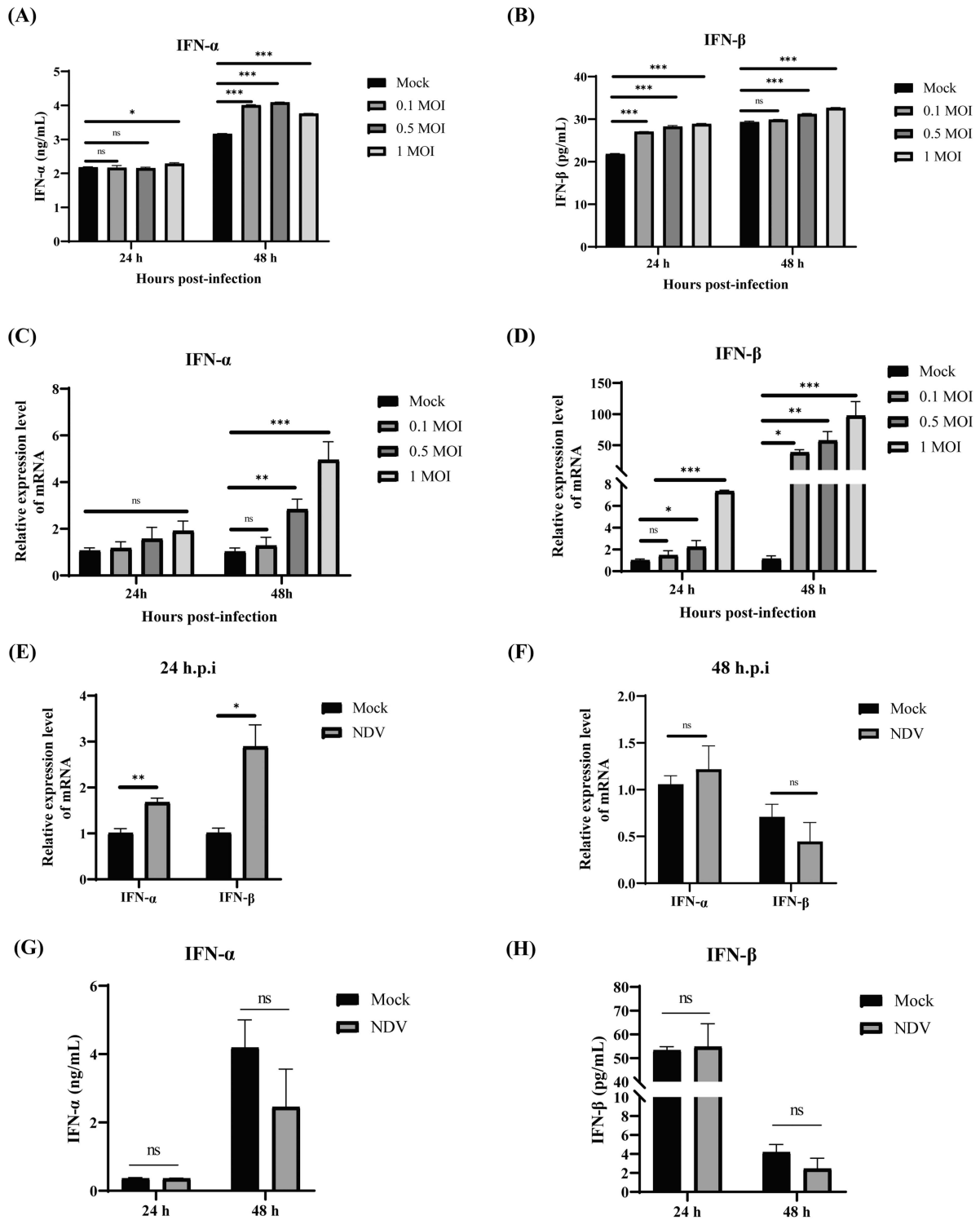


Figure 1. NDV induces type I interferon release from CMT-U27 cells. (A) IFN- α content in cell supernatants of CMT-U27 cells after infection with different titre of NDV for 24 h and 48 h; (B) IFN- β content in cell supernatants of CMT-U27 cells after infection with different titre of NDV for 24 h and 48 h; (C) Expression levels of IFN- α gene in CMT-U27 cells after infection with different titre of NDV for 24 h and 48 h; (D) Expression levels of IFN- β gene in CMT-U27 cells after infection with different titre of NDV for 24 h and 48 h; (E) Expression levels of IFN- α and IFN- β genes in MDCK cells after infection with NDV (MOI = 1) for 24 h; (F) Expression levels of IFN- α and IFN- β genes in MDCK cells after infection with NDV (MOI = 1) for 48 h; (G) IFN- α content in cell supernatants of MDCK cells after infection with NDV (MOI = 1) for 24 h and 48 h; (H) IFN- β content in cell supernatants of MDCK cells after infection with NDV (MOI = 1) for 24 h and 48 h. Results were expressed as mean $\pm$ SEM. * P < 0.05, ** P < 0.01, *** P < 0.001; ns, not significant.

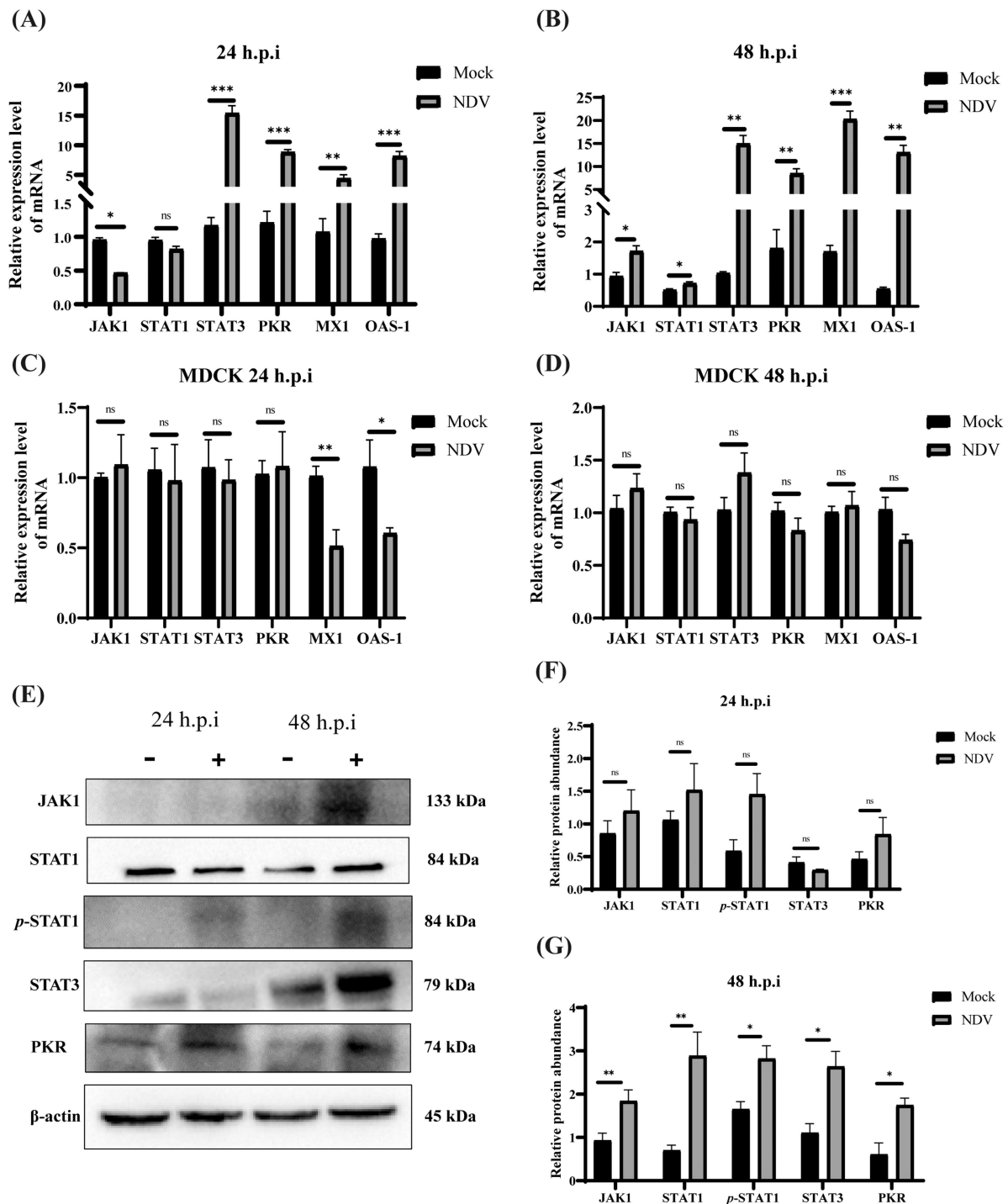


Figure 2. NDV activates the JAK-STAT signalling pathway. (A) The mRNA expression levels of the JAK-STAT signalling pathway related genes post 24 h of CMT-U27 infection with NDV (MOI = 1); (B) The mRNA expression levels of the JAK-STAT signalling pathway related genes post 48 h of CMT-U27 infection with NDV (MOI = 1); (C) The mRNA expression levels of the JAK-STAT signalling pathway related genes post 24 h of MDCK infection with NDV (MOI = 1); (D) The mRNA expression levels of the JAK-STAT signalling pathway related genes post 48 h of MDCK infection with NDV (MOI = 1); (E) The expression levels of JAK-STAT signalling pathway related proteins post 24 h and 48 h of CMT-U27 infection with NDV (MOI = 1). The samples were derived from the same experiment; (F) Quantitative analysis of proteins expression in cells 24 h.p.i.; (G) Quantitative analysis of proteins expression in cells 48 h.p.i. Results were expressed as mean $\pm$ SEM. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$; ns, not significant.

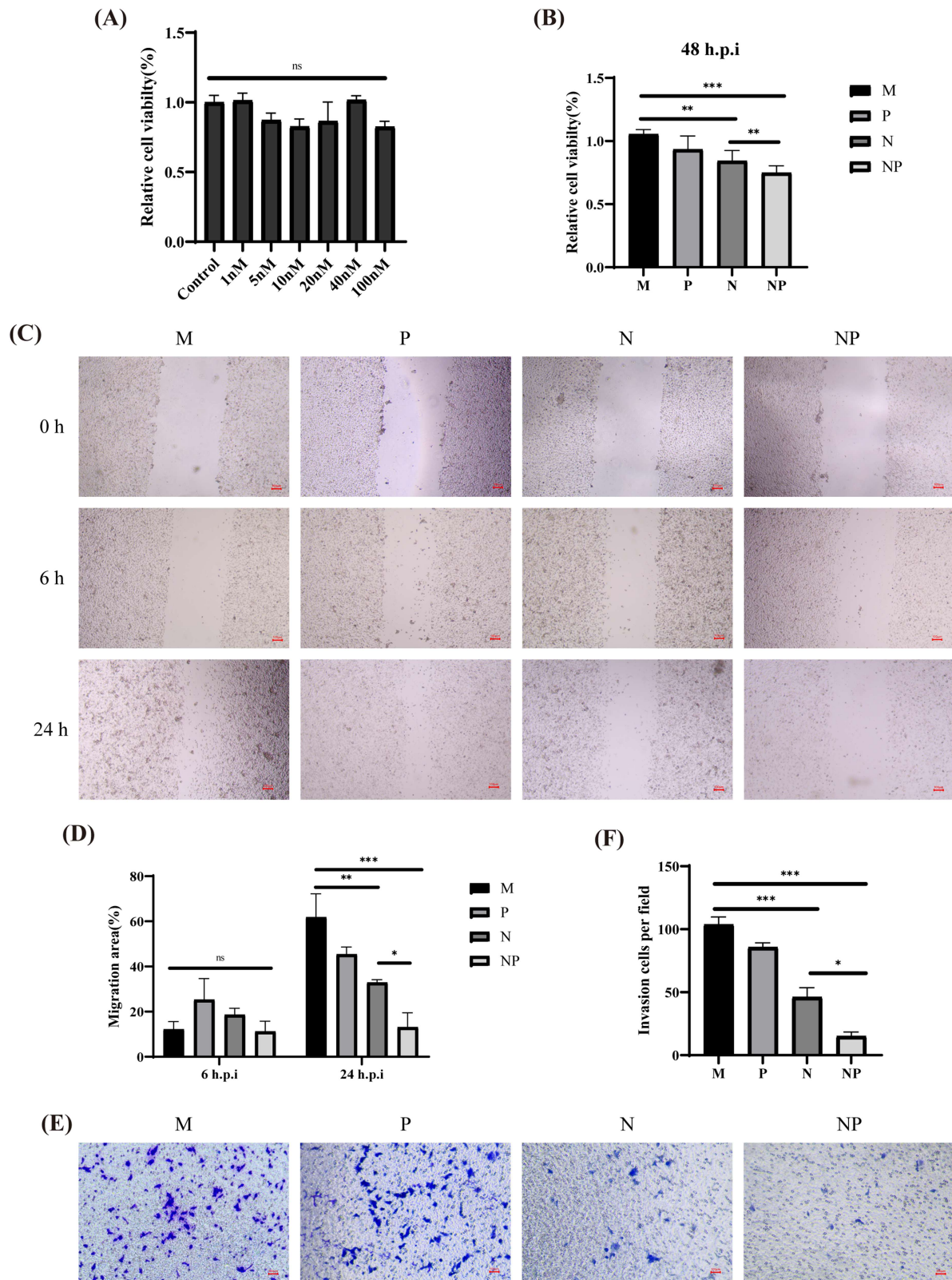


Figure 3. Propranolol enhances the inhibition of NDV on the viability, migration and invasive ability of CMT-U27 cells. (A) Cell viability of CMT-U27 cells treated with varying concentrations of propranolol for 48 h; (B) Cell viability of CMT-U27 cells treated with 100 nM propranolol in combination with NDV (1 MOI) for 48 h; (C) Migration of CMT-U27 cells at different time points after treatment with NDV (MOI=0.01) and 100 nM propranolol (scale bar=100 μ m); (D) Quantification of migration area percentage; (E) Invasion assay of CMT-U27 cells treated with NDV (MOI=1) and 100 nM propranolol (scale bar=100 μ m); (F) Number of invaded CMT-U27 cells per field. Results were expressed as mean $\pm$ SEM. * P < 0.05, ** P < 0.01, *** P < 0.001; ns, not significant.

of NDV on CMT-U27 cell viability, migration and invasion; however, according to Bliss analysis, the interaction is additive for viability and migration but synergistic for invasion under the tested condition.

3.4. Propranolol enhances NDV replication in CMT-U27 cells by inhibiting the antiviral response

Immunofluorescence analysis revealed markedly stronger NDV-HN protein signals in the NP group compared to the N group after 24 h of infection (MOI = 1), as indicated by higher integrated red fluorescence density (Figure 4A, B). These findings suggest that propranolol enhances NDV replication in CMT-U27 cells.

To determine whether propranolol influences the IFN-I response induced by NDV, we measured IFN-I production. At 48 h.p.i, propranolol alone markedly increased the secretion of IFN- α and IFN- β compared with the M group ($P < 0.001$). In contrast, the NP group displayed significantly lower levels of IFN- α and IFN- β compared to NDV infection alone ($P < 0.001$, Figure 4C, D). Consistently, mRNA expression of IFN- α and IFN- β was also markedly downregulated in the NP group relative to the N group ($P < 0.001$, Figure 4E).

Consistent with these findings, RT-qPCR and western blot analyses showed that the expression levels of JAK-STAT pathway-related genes and proteins downstream of IFN-I signalling were downregulated in the NP group compared with the N group (Figure 4F–H). These results suggest that propranolol enhances NDV replication in CMT-U27 cells by modulating the IFN-I-mediated JAK-STAT signalling pathway.

3.5. Propranolol combined with NDV promotes viral replication in vivo and inhibits tumour growth in nude mice

Kaplan-Meier survival analysis revealed that the P, N, and NP groups significantly prolonged survival compared with the M group. All nude mice in the M group died by day 11 post-injection, while all mice in the propranolol group died by day 14. In contrast, two mice in the N group and three nude mice in the NP group survived until day 15. However, no statistically significant difference in survival was observed between the N and NP groups (Figure 5A).

Body weight monitoring revealed a marked decline in the M group from 16.74 ± 0.65 g to 12.84 ± 0.38 g by day 10 ($P < 0.01$), while no significant weight loss was observed in the P, N or NP groups during the same period (Figure 5B).

Relative to the M group (640.15 ± 201.58 mm³), mean tumour volumes on day 10 were significantly lower in both the N (319.44 ± 154.71 mm³; $P < 0.05$) and NP (245.60 ± 63.40 mm³; $P < 0.01$) groups. The P group showed no significant reduction in tumour volume (630.29 ± 168.47 mm³). No statistically significant difference was observed between the N and NP groups (Figure 5C, D). Furthermore, immunofluorescence analysis revealed stronger NDV-HN protein expression in tumour tissues from the NP group compared with the N group ($P < 0.05$), indicating enhanced viral replication *in vivo* (Figure 5E, F).

3.6. Propranolol inhibits NDV-activated antiviral responses in nude mice

Serum IFN-I levels were elevated in NDV-treated mice compared with the M group ($P < 0.05$). Although propranolol co-administration showed a decreasing trend in the serum IFN-I levels, only the reduction in IFN- β reached statistical significance compared with N group ($P < 0.05$) (Figure 6A, B).

Consistently, analysis of tumour tissues revealed that NDV infection significantly upregulated the mRNA expression of JAK1, STAT1, STAT3, PKR and MX1 compared with the M group ($P < 0.05$). In contrast, the NP group showed markedly reduced expression of these genes relative to the N group (Figure 6C).

At the protein levels, NDV treatment strongly induced the expression of JAK1, p-JAK1, STAT1, p-STAT1, STAT3 and PKR ($P < 0.05$) whereas co-treatment with propranolol significantly inhibited the expression of these proteins, except for STAT1, whose suppression did not reach statistical significance (Figure 6D, E).

4. Discussion

Previous studies have shown that oncolytic viruses such as NDV can induce tumour-specific hyperactivation of IFN-I/III pathway, leading to selective tumour cell clearance through differential activation of apoptotic signalling pathways (Ginting and Christian, 2019). In line with these findings, our study demonstrated that

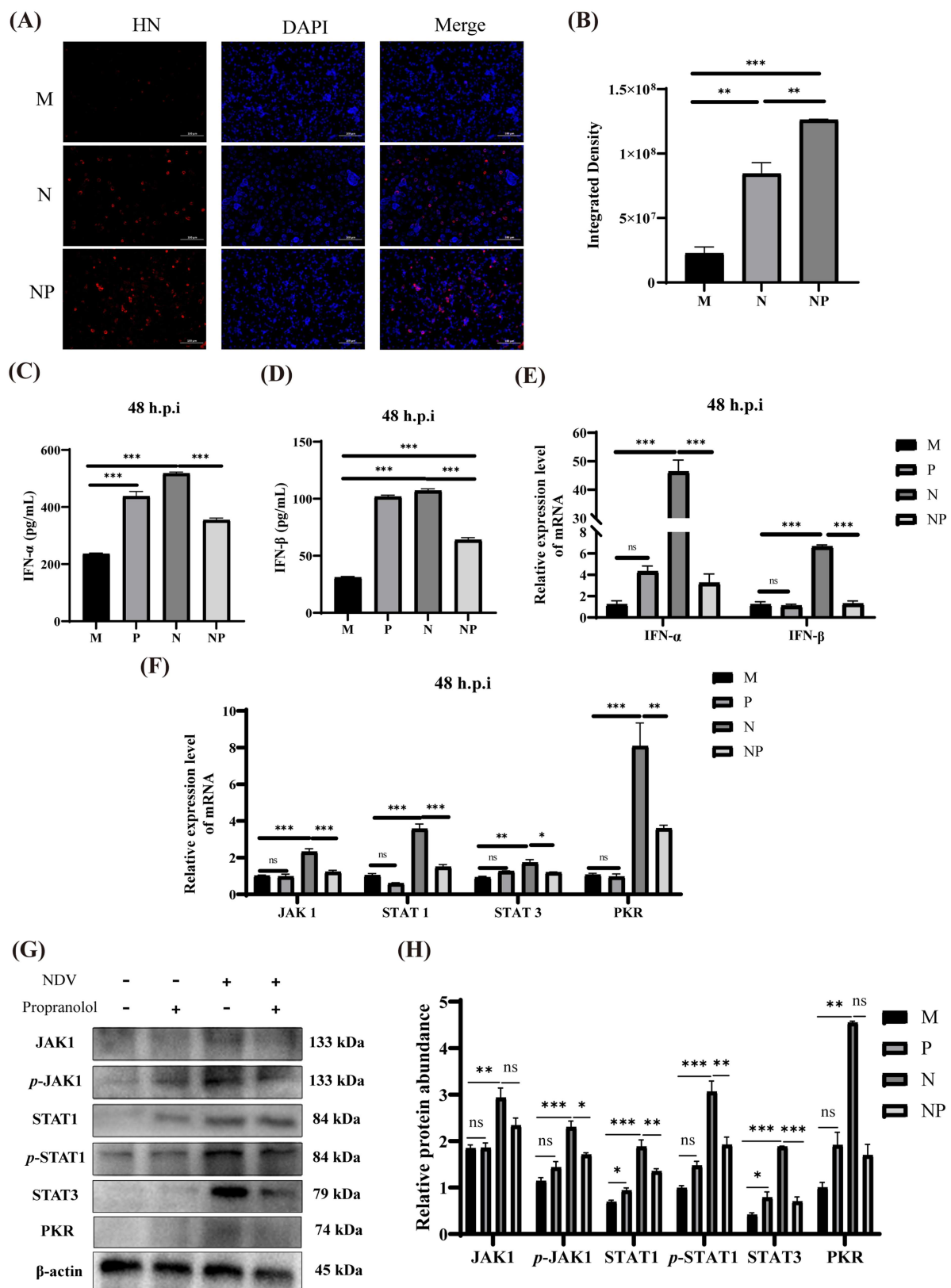


Figure 4. Propranolol enhances NDV replication in CMT-U27 cells by inhibiting the antiviral response in these cells. (A) Immunofluorescence analysis of NDV-HN protein expression in CMT-U27 cells after treatment with NDV (MOI = 1) alone or in combination with 100 nM propranolol for 24 h (scale bar = 100 μ m); (B) Quantification of NDV-HN fluorescence intensity in each group; (C) Release of IFN- α after 48 h treatment of CMT-U27 cells with propranolol (100 nM) and NDV (1 MOI) alone or in combination; (D) Release of IFN- β after 48 h treatment of CMT-U27 cells with propranolol (100 nM) and

(Caption on next page)

NDV (1 MOI) alone or in combination; (E) Expression level of IFN-I gene mRNA in CMT-U27 cells after 48 h treatment of propranolol (100 nM) and NDV (1 MOI) alone or in combined treatment for 48 h; (F) mRNA expression levels of JAK-STAT signalling pathway-related genes in CMT-U27 cells treated with propranolol (100 nM) and NDV (1 MOI) alone or in combination for 48 h; (G) Protein expression levels of JAK-STAT signalling pathway-related proteins in CMT-U27 cells treated with propranolol (100 nM) and NDV (1 MOI) alone or in combination for 48 h. The samples were derived from the same experiment; (H) Quantitative analysis of protein levels after 48 h of treatment. Results were expressed as mean $\pm$ SEM. * P < 0.05, ** P < 0.01, *** P < 0.001; ns, not significant.

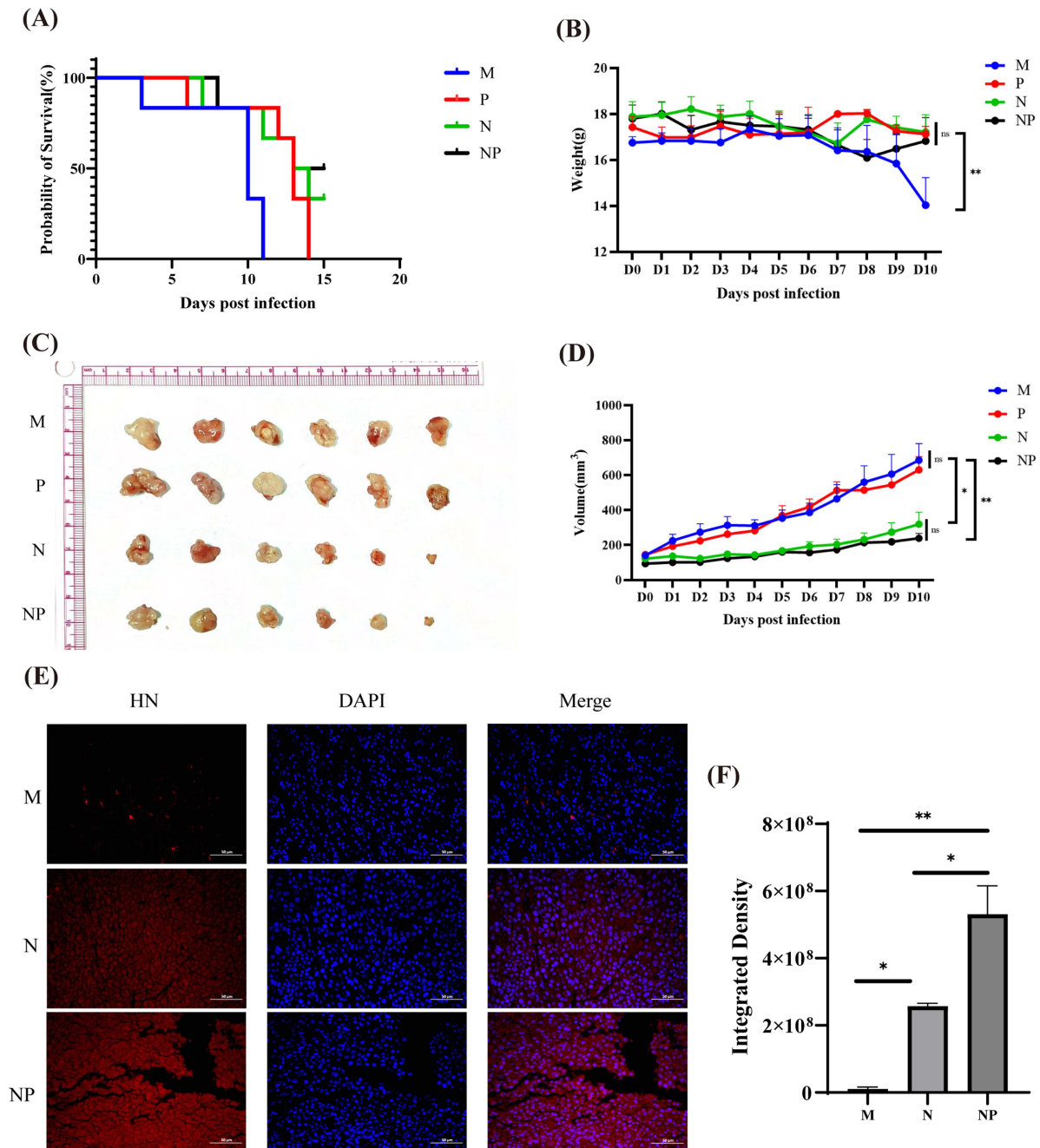


Figure 5. Propranolol combined with NDV promotes viral replication *in vivo* and inhibits tumour growth in nude mice. (A) Kaplan-Meier survival curve of tumour-bearing nude mice during treatment ($n = 6$); (B) Body weight of tumour-bearing nude mice during treatment ($n = 6$); (C) Comparison of tumour size of tumour-bearing nude mice in different treatment groups ($n = 6$); (D) Tumour growth in tumour-bearing nude mice during treatment ($n = 6$); (E) Expression of NDV-HN protein in tumour tissues of nude mice in each treatment group (scale bar = 50 μ m); (F) Quantification of NDV-HN fluorescence intensity in each group of tumour tissues ($n = 3$). Results were expressed as mean $\pm$ SEM. * P < 0.05, ** P < 0.01, *** P < 0.001; ns, not significant.

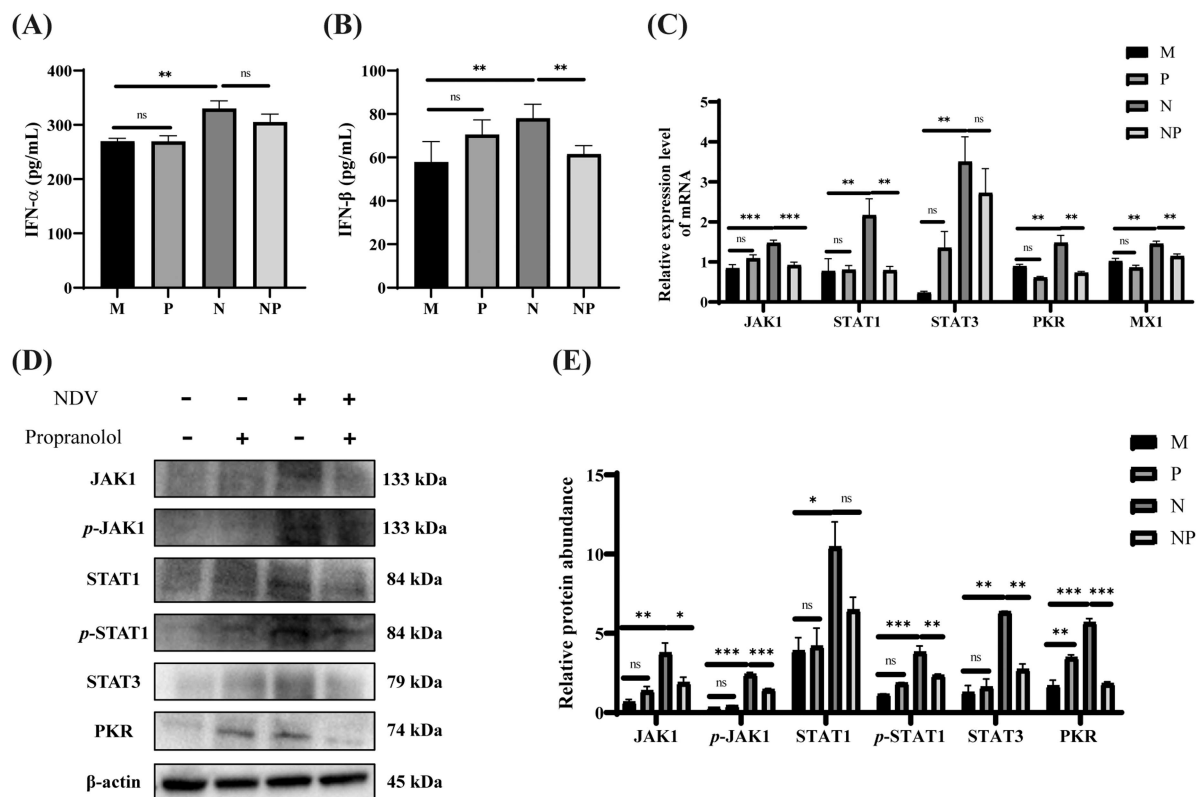


Figure 6. Inhibition of NDV-activated antiviral responses in nude mice by propranolol. (A) IFN- α content in the serum of tumour-bearing nude mice ($n = 3$); (B) IFN- β content in the serum of tumour-bearing nude mice ($n = 3$); (C) The expression levels of mRNA related to the JAK-STAT signalling pathway in tumour tissues of nude mice across various treatment groups ($n = 3$). The samples were derived from the same experiment; (D) Expression of JAK-STAT signalling pathway-related proteins in each treatment group ($n = 3$); (E) Quantitative analysis of tissue proteins ($n = 3$). Results were expressed as mean $\pm$ SEM. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$; ns, not significant.

NDV infection promotes the expression and release of IFN-I in CMT-U27 cells in a dose- and time-dependent manner. NDV can also significantly increase the expression of IFN-I in the CMT xenograft mouse model. Notably, prior studies observed reduced NDV-HN protein expression in CMT-U27 cells compared to MDCK cells under same conditions, indicating lower permissiveness of MDCK cells to NDV infection (Wang and Li, 2023). In this experiment, we found no NDV-induced IFN-I release or gene expression in MDCK cells at 48 h.p.i. This absence of response likely reflects the low susceptibility of normal tissue cells to NDV infection. These findings strongly confirm the tumour-specific IFN-I-dependent apoptosis (Ginting and Suryatenggara, 2017).

However, IFN-I plays a dual role in oncolytic therapy. While it enhances antitumor immunity by activating recognition receptors (PRRs), triggering pro-inflammatory cytokine release, remodelling the tumour micro-environment (TME) and boosting immune responses (Li and Wu, 2021; Zhao et al., 2020). It also activates antiviral defenses via the JAK-STAT signalling pathway, which induces interferon-stimulated genes that regulate cell proliferation, apoptosis, and antiviral responses, potentially limiting the efficacy of oncolytic viruses (Hui et al. 2014). For instance, IL-6-mediated STAT3 hyperactivation can suppress innate immune pathways like cGAS-STING, thereby reducing viral potency (Basu et al. 2006). In this study, RT-qPCR, and western blot analyses demonstrated that NDV infection activates JAK-STAT-mediated antiviral responses both *in vitro* and *in vivo*. Conversely, inhibition of JAK-STAT signalling (e.g. via SOCS3-mediated negative regulation (Yan et al. 2024) can weaken antiviral defenses, promoting viral replication and enhancing antitumor immunity. Additionally, IFN-I can drive therapeutic resistance by reshaping immune checkpoint molecule expression (Udayakumar et al. 2020; Chen et al., 2023), highlighting the need for dynamic models that account for viral load, immune activation intensity, and tumour heterogeneity. Therefore, modulating IFN release and JAK-STAT signalling may offer a promising therapeutic strategy.

Propranolol, a non-selective β -adrenergic receptor antagonist widely used in cardiovascular treatment, has emerged as a potential adjuvant in cancer therapy by disrupting catecholamine signalling within the TME and inhibiting tumour growth and metastasis (Cole and Sood 2012; Solernó et al., 2022). It exerts antitumor effects through multiple mechanisms, including induction of apoptosis and autophagy, G0/G1 phase cell cycle arrest, blockade of β -adrenergic signalling, and inhibition of the cAMP/PKA pathway, thereby directly suppressing tumour cell proliferation (Gales et al. 2022; Di Fonte et al., 2023). In addition, propranolol inhibits angiogenesis, remodels the TME by suppressing the activation of tumour-associated macrophages, and enhances antitumor immune responses (Albiñana et al. 2022). Collectively, these multifaceted effects may contribute to suppress tumour progression and improve therapeutic sensitivity.

However, the role of propranolol in NDV-infected tumour cells remains unclear. We hypothesise that the underlying mechanism may involve IFN and its downstream pathways. A previous study reported that propranolol significantly stimulated the STAT3/SOCS3 signalling axis in spinal cord microglia and myeloid cells, downregulated the expression of chemokines (CCL2, CCL19/21), promoted T cell and monocyte trafficking to the spinal cord, and reduced the number of CD4⁺IL17⁺ cells co-expressing IFN- γ (Pilipović and Stojić-Vukanić, 2023). Another study showed that propranolol modulates the immune response to cancer by reducing IFN- γ production, thereby attenuating IFN- γ -mediated PD-L1 overexpression (Falcinelli et al. 2023). In addition to its effect on IFN expression, modulation of the JAK-STAT pathway also influences tumour cell sensitivity to therapy. For example, the JAK-STAT inhibitor ruxolitinib has been shown to reverse tumour resistance to Reolysin by blocking JAK-STAT pathway activation induced by human T-cell leukaemia virus type 1, thereby enhancing Reolysin's therapeutic efficacy (Islam et al. 2021). These findings suggest that suppressing IFN signalling and its downstream pathways may improve antitumor efficacy. In the present study, we demonstrate that combined propranolol and NDV treatment suppresses the IFN-I-mediated JAK-STAT signalling pathway and enhances NDV replication both *in vivo* and *in vitro* compared with NDV monotherapy. Furthermore, propranolol enhanced NDV-induced inhibition of CMT-U27 cell viability, migration, and invasion, and enhanced NDV replication while suppressing tumour progression in xenograft models.

The combination of propranolol with chemotherapeutic agents or oncolytic viruses has emerged as a promising therapy for cancer treatment. Alzahrani et al. demonstrated that propranolol synergises with capecitabine to induce ferroptosis in human colorectal cancer cells, highlighting its potential as an adjuvant to capecitabine in colorectal cancer therapy (Alzahrani et al. 2025). Hu et al. reported that propranolol in combination with the oncolytic virus T1012G exerts synergistic cytotoxic effects in gastric and colorectal cancer models (Hu et al., 2021). In the present study, we extend this concept to CMTs, showing that propranolol combined with NDV modulates IFN-mediated antitumor immune and antiviral responses, thereby enhancing NDV's oncolytic efficacy. Our findings suggest that propranolol-NDV combination therapy represents a novel and promising strategy for the treatment of CMTs. Although the combination of propranolol and NDV significantly enhanced viral replication and suppressed tumour cell malignancy *in vitro*, this did not translate into a statistically significant improvement in tumour growth inhibition or survival in the CMT-U27 xenograft model compared to NDV monotherapy. This discrepancy between *in vitro* and *in vivo* outcomes warrants careful consideration.

Several factors may explain this observation. First, the *in vivo* TME is considerably more complex than monoculture systems, and may modulate the effects of both NDV and propranolol. Stromal components, immune infiltrates, and hypoxic conditions within the TME could attenuate viral spread or dampen the synergistic effects observed *in vitro* (Thomas et al. 2025). Second, the dosing regimen and route of propranolol administration may not have been optimal for maintaining intratumoral concentrations sufficient to continuously suppress the IFN-I-mediated antiviral response. Although propranolol was administered intraperitoneally at a dose previously shown to be effective in other tumour models (Zheng and Du, 2024), its pharmacokinetics and tissue distribution in nude mice bearing CMT-U27 tumours remain unclear. Third, the relatively short treatment duration combined with the aggressive growth of CMT-U27 tumours may have limited the window for detecting divergence between the N and NP groups. Earlier initiation or prolonged dosing schedules might reveal greater therapeutic benefits. Moreover, while propranolol enhanced NDV replication within tumours, this did not translate into greater tumour regression, suggesting that viral load is insufficient to determine oncolytic efficacy. Additional factors such as tumour cell susceptibility to virus-induced cell death, immune-mediated clearance, and the spatial distribution of viral infection within the tumour mass likely contribute to the overall therapeutic outcome (Schoeps et al. 2025).

To address these challenges, several strategies warrant consideration. These include optimising the dose and schedule of propranolol to achieve sustained modulation of the JAK-STAT pathway (Yamada et al., 2025); employing orthotopic or spontaneously occurring CMT models that better recapitulate the TME and host immune interactions (Stribbling et al. 2024); incorporating immune-competent models to evaluate the contribution of adaptive immunity to combination therapy (Clubb et al. 2023); and exploring alternative delivery systems, such as intratumoral injections of propranolol-loaded nanoparticles, to achieve localised and sustained drug release (Jiang and Fu , 2024). Future investigations are also needed to assess a broader range of propranolol-NDV dosing regimens and to confirm these findings in additional CMT cell lines. In addition, since MDCK cells do not fully recapitulate the physiological characteristics of normal canine mammary epithelial cells, more representative control models are required.

5. Conclusion

In conclusion, this study demonstrates that NDV induces IFN-I release and activates the JAK-STAT signalling pathway in CMT-U27 cells. Propranolol enhances NDV replication and oncolytic efficacy in CMT-U27 cells by suppressing the antiviral response. Although the combination therapy did not confer additional survival or tumour growth benefits compared with NDV monotherapy in the CMT-U27 xenograft model, it significantly increased viral load within tumour tissues, highlighting its capacity to improve the intratumoral efficacy of NDV. Collectively, these results provide mechanistic insight into the interaction between β -adrenergic signalling and antiviral responses and support the potential of propranolol as an adjuvant to optimise NDV-based virotherapy.

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Author contributions

YZ: Writing - original draft, Conceptualisation, Data curation, Formal analysis, Methodology, Project administration, Visualisation; MX: Writing - review & editing, Data curation, Formal analysis, Project administration, Visualisation; ML: Writing - review & editing, Conceptualisation, Funding acquisition, Project administration, Supervision, Visualisation.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The original contributions presented in the study are publicly available. This data can be found here: <http://doi.org/10.17632/nf6843cxgr.1>

Ethics approval and consent to participate

The study was approved by the Institutional Animal Care and Use Committee of Nanjing Agricultural University (No. 20241015206).

Consent for publication

All authors approved the manuscript and gave their consent for submission and publication.

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